11 Control Systems with Electrical Fields

Despite the ubiquitous use of electrical stimulation there have been few basic studies of the biophysics of the process.
—Tranchina and Nicholson, 1986 [TN86]

11.1 Introduction

Most neurophysiologists have an affair with electrical fields. For some, such as the author, the infatuation starts early and never abates. My guess is that reading Hodgkin and Huxley's papers at an impressionable age may be responsible. It can lead to obsession.

If we assimilate data from the nervous system, what can be done with such information has been the subject of considerable discussion. Some suggest ringing little alarm bells from a device to tell a person that something bad is about to happen in their nervous system; this could allow you to park your car at the side of the highway, for example, before you have an epileptic seizure—the electronic analog of a service dog. Another alternative is to inject drugs into part of a nervous system in response to signals from a data assimilation system—not the worst idea, especially if you want those drugs to target a particular place in the brain to modify a disease state such as epilepsy or Parkinsons disease, as opposed to bathing the whole brain or body in the medication. Most of us gravitate toward electrical stimulation. Such stimulation can be delivered by induction through changing magnetic fields, or applied directly as quasi-static electrical fields (ignoring induced magnetic fields) [NS06].

11.2 A Brief History of the Science of Electrical Fields and Neurons

Digging into the history of electrical fields fairly rapidly drags the reader back to the eighteenth-century experiments of Luigi Galvani and his wife Lucia Galeazzi [Bar89, Pic97]. In experiments in their backyard, they connected an exposed frog spinal cord to a wire leading to their house's lightning rod, and grounded the frog's leg to their drinking well. When lightning struck nearby, the electrical field developed sufficient potential in their circuit to cause the frog leg to twitch (figure 11.1).

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

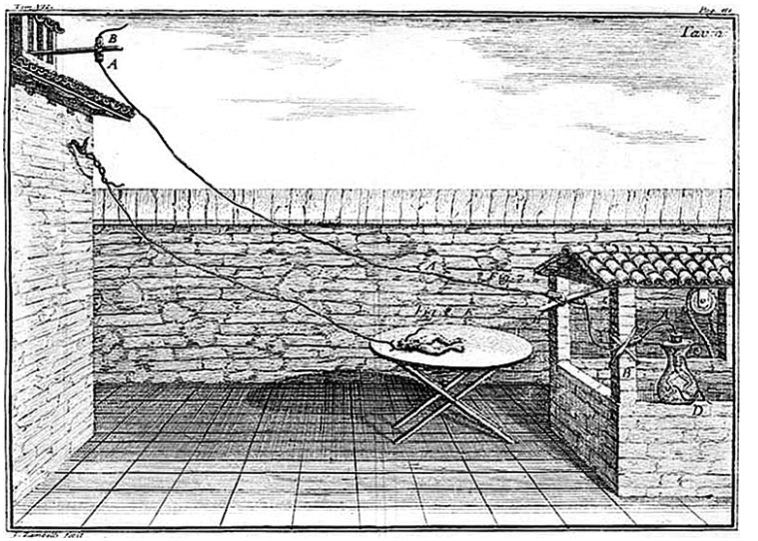


Figure 11.1
Galvani's plate from his 1791 text [Gal91]. From the facsimile generously made available from the efforts of the International Centre for the History of Universities and Science, Department of Philosophy, University of Bologna.

By the early twentieth century, William Rushton performed the seminal electrical field experiment that lays out most of what the researcher would need to replicate the rest of this chapter [Rus27]. He knew of Galvani's work, and fashioned an ingenious experiment. He suspended a frog sciatic nerve between two insulating glass receptacles. The nerve could be stimulated within one insulator, and the action potentials then traveled along the length of the nerve suspended in a bath of Ringer's solution. The other end of the nerve was attached to the calf muscle, which would twitch if the impulse arrived. The nerve was suspended between two silver-silver chloride plates, which generated a uniform electrical field in the region of the nerve. The apparatus was placed on a turntable, graduated in degrees, so that he could change the angle of the nerve with respect to the electrical field (figure 11.2). He found that the threshold of current required, using a short square wave stimulus created through the swinging of a pendulum, was related to the cosine of the angle of the electric field, with respect to the long axis of the nerve.

Rushton's work was known to Bernard Katz and Otto Schmitt [KS40], who demonstrated that action potentials in one nerve fiber could influence the transmission of action potentials in a neighboring fiber. The physics of such influence would serve to synchronize nearby action potentials so that they traveled together. These effects were termed *ephaptic*.[1]

1. It goes without saying that Katz, who would later receive the Nobel Prize in medicine and physiology, was fortunate in collaborating with the young Schmitt. Schmitt would later become one of the most important electronics inventors of the twentieth century. His *Schmitt Trigger* was directly inspired by the knowledge he gained from experiments on all-or-none action potentials in nerve fibers.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

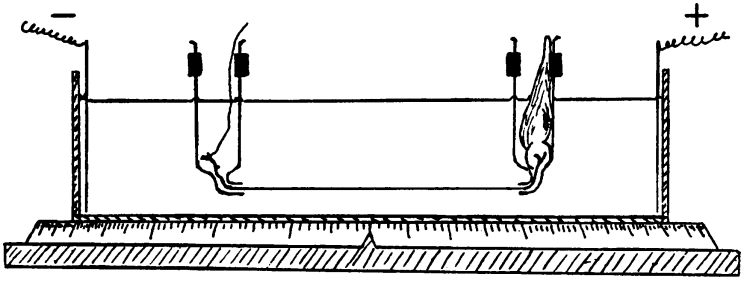


Figure 11.2
Rushton's experiment exploring the relationship between the threshold of a nerve fiber and the angle with respect to electrical field. Reproduced from [Rus27] with permission.

By 1956, Terzoulo and Bullock [TB56] demonstrated that the electrical field required to modulate an active neuron was far lower than that required to stimulate one to fire from rest. Remarkably, they estimated that the strength of the electrical field required to perform such modulation was of the order of 1 mV/mm, an estimate not to be broken by theory [WVAA98] or experiment [FGS03] for over four decades. It was clear to Terzuolo and Bullock [TB56] that these magnitudes of fields were "available" to neurons within active nervous systems, and therefore might play a role in synchronization. They further recognized that the morphology of such cells would play a role in their sensitivity to electrical fields, that their dendrite fields would act as *antennae* when oriented parallel to the field direction, and that the arrangement of neurons in mammalian cortex were especially set up with such alignments.[2]

By the late 1970s, we developed the ability to cut thin slices of brain and keep such slices alive for many hours in perfusion chambers [And77]. The living and physiologically active brain slice was soon viewed as an ideal new window to begin exploring the effects of electrical fields on neuronal and network activity [Jef81].

In a bizarre set of experiments in the early 1980s, researchers noted that neurons could form organized activity and synchronize in the absence of chemical transmission in brain slices [JH82, TD82, YKH83, KHY84]. In a series of computational modeling studies, Traub and colleagues [TDTK85, TDSK85] explored the origin of such pathological synchronization as well as the implications for more physiological synchronization of individual neurons to populations.

But the remaining piece of this puzzle was to be worked out by Charles Nicholson and his colleagues, in a remarkable set of experiments in the mid-1980s. Using isolated turtle cerebellum in an elegant experimental setup, they demonstrated in neuronal populations that the asymmetry of the soma-dendritic axis correlated with the responsiveness of such cellular populations to modulate their firing in response to relatively slow sinusoidal stimulation

2. Many of us will sadly mark the recent loss of Ted Bullock. I was deeply impressed by the interest that this preeminent scientist took in inspiring school students.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

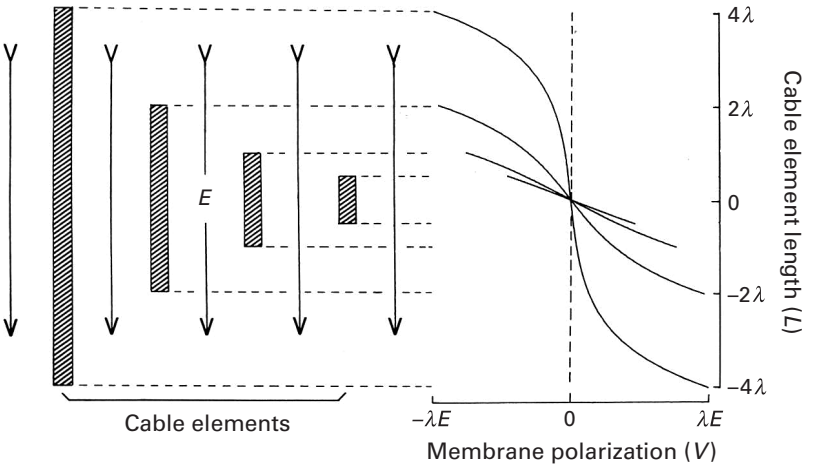


Figure 11.3
Semileaky cables exposed to a constant electrical field will have sigmoid-shaped (hyperbolic tangents) voltage profiles. Reproduced from [CN86] with permission.

[CN86]. They proposed that from passive cable calculations[3] [Ran75] the potential profile, $V(x)$, would vary with distance x as a sigmoid (hyperbolic tangent):

$$V = \lambda E \sinh(x/\lambda - L/\lambda)/\cosh(L/\lambda)$$

where λ is a space constant, E the strength of the constant polarizing electrical field, $2L$ the length of the cable, and x the distance along the cable (figure 11.3). Two years later, Chan and colleagues [CHN88] demonstrated from intracellular measurements that such field effects on neuronal spiking were consistent with these cable equations.[4]

But embodying all of this experimental science was the seminal computational work by David Tranchina and Nicholson [TN86]. In this work, they considered that, since the dendritic branches in a neuron could be electrically equivalent to a simpler structure, they created an equivalent cylinder to account for the soma and dendrite fields, as shown in figure 11.4.

It is important to realize that whereas the gradient (first derivative) of the potential (the electrical field) is the crucial element in polarizing the equivalent cylinder in figure 11.4, it

3. Such cable equations are actually derivative of the cable equations for the nineteenth-century transatlantic cables.

4. It is worth noting that there was an older literature demonstrating opposite effects of DC polarization of neurons embedded within the intact brain, cotex [PM65] and hippocampus [PM66], which these new findings now explained. For large asymmetric principal (pyramidal) cells, the orientation of the large apical dendrite with respect to the soma was the key to predicting whether polarization would be excitatory or inhibitory.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

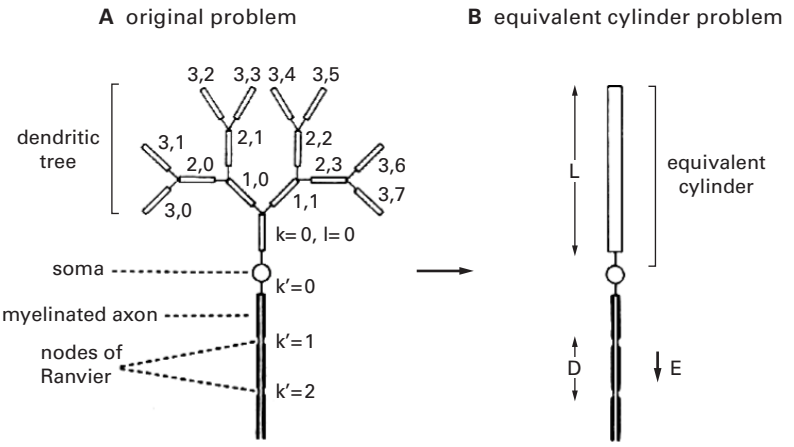


Figure 11.4
Converting a neuron with a complex dendritic tree in A, to the equivalent cylinder model of dendrite, soma, and axon in B. Reproduced from [TN86] with permission.

is the second derivative of the potential[5] that is important if you wish to activate the middle of a cylindrical structure [WGD92] (see also related discussion in chapter 10).

A few more facts need to be mentioned in this too-brief summary. The time constant to polarize a neuron in the way we have been discussing is about 25 msec [BIA+04]. This corresponds to maximal frequencies of about 20–40 Hz.[6] Interestingly, higher-frequency sinusoidal stimulation will induce a high potassium outflow that can block neuronal activity (depolarization block), and is orientation independent [BLH+01].

So we are done with the basic science of this chapter. Much of it can be summarized in the extremely simple diagram in figure 11.5.

11.3 Applications of Electrical Fields in Vitro

Our studies also indicate the utility of this relatively simple stimulation method that enables populations of cells to be stimulated without penetration of individual cells. Moreover the current density is quite low at any point in the tissue since the current is not delivered by a local punctate electrode . . . Such a stimulus is potentially valuable in prosthetic devices.
—Chan and Nicholson, 1986 [CN86]

This book is concerned with pushing the limits of neural control engineering. The vast majority of applications of stimulation in both experimental and clinical nervous system

5. For better or worse, this second derivative is often termed an *activating function* in the literature.

6. There is evidence that synaptic afferents might analogously polarize [BIA+04].

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

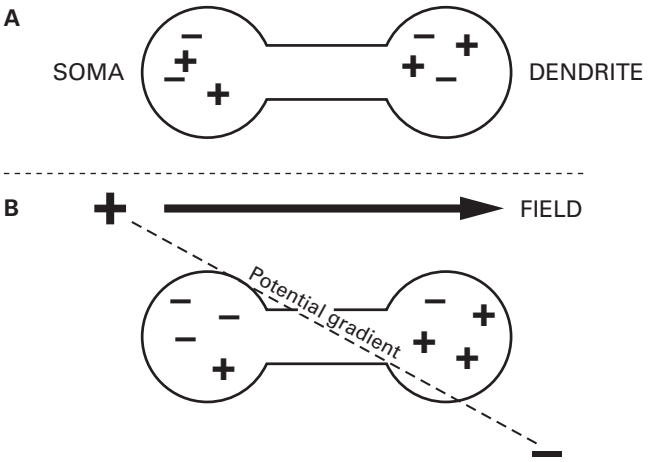


Figure 11.5
The completely reduced view of low-frequency electrical field polarization of neurons along their soma-dendritic axes. A shows the situation without an applied field. Electroneutrality resides along the axial direction of the neuron. In B, with steady polarization along the soma-dendritic axis, there is a small separation of charges, which actually build up along the membranes at the ends of the neuron as a Debye layer. This produces a counterfield (opposite in direction) to the applied field. Since the transmembrane (inside versus outside) potential difference near the soma determines the threshold of the neuron to fire, the polarization of such a neuron will make it either more or less excitable, depending on whether the potential near the soma is hyperpolarized (less excitable) or depolarized (more excitable).

applications are from pulse stimulation. In such scenarios, one delivers a *square wave* pulse of either voltage or current, depending on the source of energy. There are advantages and disadvantages of each. If one delivers constant voltage, the junction of the electrodes with neural tissue can vary widely, and thus the current delivered can similarly be unreliable. So in many experimental situations, constant current produces more reliable results. On the other hand, if one exceeds certain voltage limits, one can readily form irreversible reduction-oxidation reactions at the electrode surface with neural tissue [MBJ05], so constant voltage is safer.

But the problem with square wave–type pulse stimulation is that the rapidly changing electrical field associated with such stimuli, which turn on and off with a theoretically "infinite" rate of change in voltage or current, is that the ability to simultaneously measure the response of the neural activity is lost. One can, of course, record simultaneously, but we have no way of accurately separating the stimulation artifact from neural activity during such rapid shifts in stimulation energy.

The Chan and Nicholson quotation from 1986 at the start of this section was one of the least noticed paragraphs in neuroscience for about a decade. After nailing down the advantages of low-frequency electrical fields for neural prosthetic applications using extracellular, intracellular, and computational models, their work lay in wait for the rest of us to study. So

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

I am going to pick up on this thread for the rest of this chapter. This is not to say in any way that the usefulness of pulsed stimulation is not well established in many applications. Certainly we think it is required in areas of the brain, such as stimulation of the basal ganglia for Parkinson's disease (chapter 10), where neuronal orientation may not permit low-frequency field stimulation. Nor do I intend to suggest that we understand pulsed stimulation within the brain—the situation is anything but clear [MGST04]. But low-frequency field stimulation has serious advantages for a control engineer, and the literature on applications is a growing area of focus. As we will show, it offers a powerful way to separate the stimulation from neural response in continuous feedback scenarios.

Bruce Gluckman built a miniature version of the Rushton chamber (without the turntable), and mapped the electrical field throughout (figure 11.6). We placed a reference electrode in the chamber outside of the slice as well as a differential electrode within the slice. By turning on a test field such as a sinusoid, and moving the reference electrode with respect to the field isopotentials, the isopotential aligned with the slice electrode is found when the test signal disappears. Once this is set, one can apply any arbitrary field and it is removed from the recording. If only this could be readily accomplished in the intact brain.

It was relatively straightforward to demonstrate that suppression of seizure-like events could be achieved with DC electrical fields aligned with the neurons firing seizures [GNN$^+$96a]. But these steady polarizations were largely temporary. After about 30 or more seconds of steady polarization, seizures would return. An interesting effect of nonorientation-specific higher-frequency sinusoidal stimulation was that the poststimulation effect on activity suppression was longer lasting—likely due to potassium extrusion from cells [BIA$^+$04].

From a control point of view, examine the results in figure 11.7. In panel A, as the applied electrical field polarity is flipped positive and negative, and with graded amplitude, the burst rate of the ensemble of neurons responds in a graded manner [GNN$^+$96a]. What about individual ensemble bursts? In figure 11.7B, we demonstrate that by randomly selecting the field polarity, amplitude, and duration, we could parametrically control the individual burst intervals (unpublished data, courtesy B. J. Gluckman).

11.4 A Brief Affair with Chaos

In 1994, we published a demonstration of chaos control in such brain networks [SJD$^+$94]. Using pulse stimulation, we could shorten the interval reliably from a previous burst until the next burst. This gave us some ability to manipulate the ratio of the interval from one interval to the next, as shown in figure 11.8 (plate 29). The reason for these findings is that in chaotic systems, although all periodicities can be unstable, all are not equal. Indeed, such systems contain a hierarchy of unstable cycles called *unstable periodic orbits* [Cvi88]. In 1990 Edward Ott, Celso Grebogi, and James Yorke proposed that such systems could be controlled through manipulating about such low-period unstable periodic orbits

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

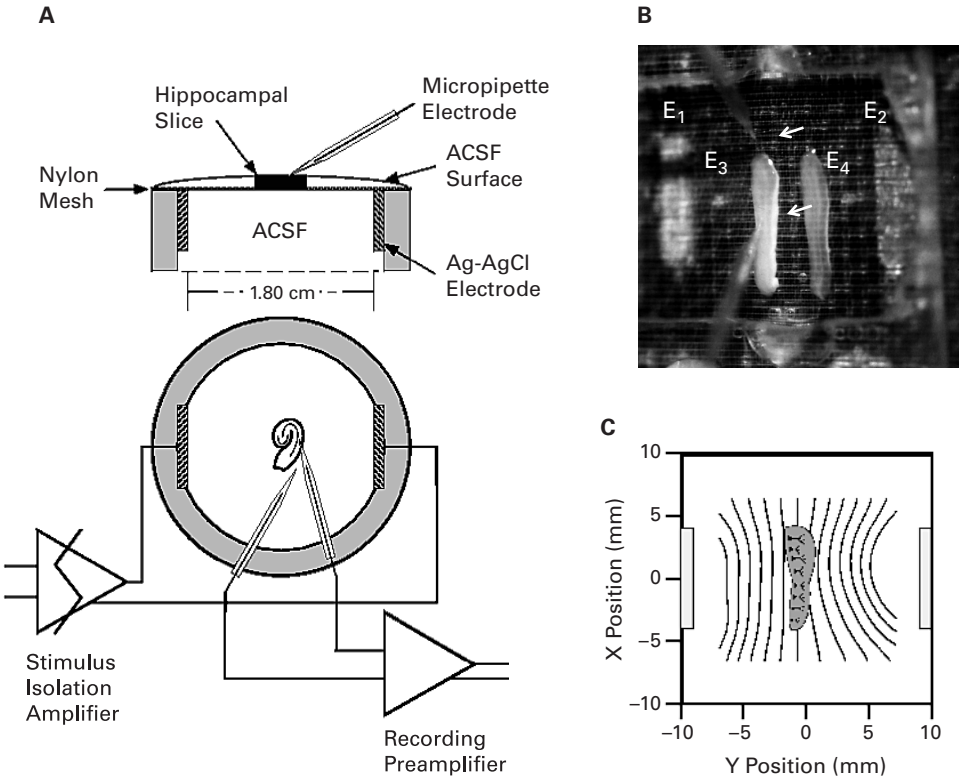


Figure 11.6
Electrical field control chamber for brain slice experiments. A shows the overall schematic; the photograph in B shows an example of two brain slices within such a chamber; and C is an actual mapping of field within the chamber. Note the presence of four electrodes in B: E_1 and E_2 generate the field through a low-source impedance current source, while the smaller E_3 and E_4 are connected to high-impedance field effect transistors for sensing the field with minimal current flow. This configuarion is a large-scale voltage clamp, with a four-electrode technique, as discussed in chapter 3. The arrows in B point to the position of the tips of the electrode in the slice and the electrode in the chamber. A and C are reproduced from [GNN+96a] with permission.

[OGY90], and William Ditto and colleagues experimentally verified this theory shortly thereafter in a physical [DRS90] and then a cardiac [GSDW92] system. But what we never could do in such control experiments was to lengthen, as well as shorten, such intervals. The primary motivation for figure 11.7 was to be able to find a true control parameter that could produce more refined control than the rather crude approach shown in figure 11.8.

The underlying idea for control of chaotic systems is actually simple. Chaos in its pure sense is an infinite number of frequencies that are *all* unstable. But there is a hierarchy of these cycles, and the low-period ones are experimentally accessible. You see them in

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

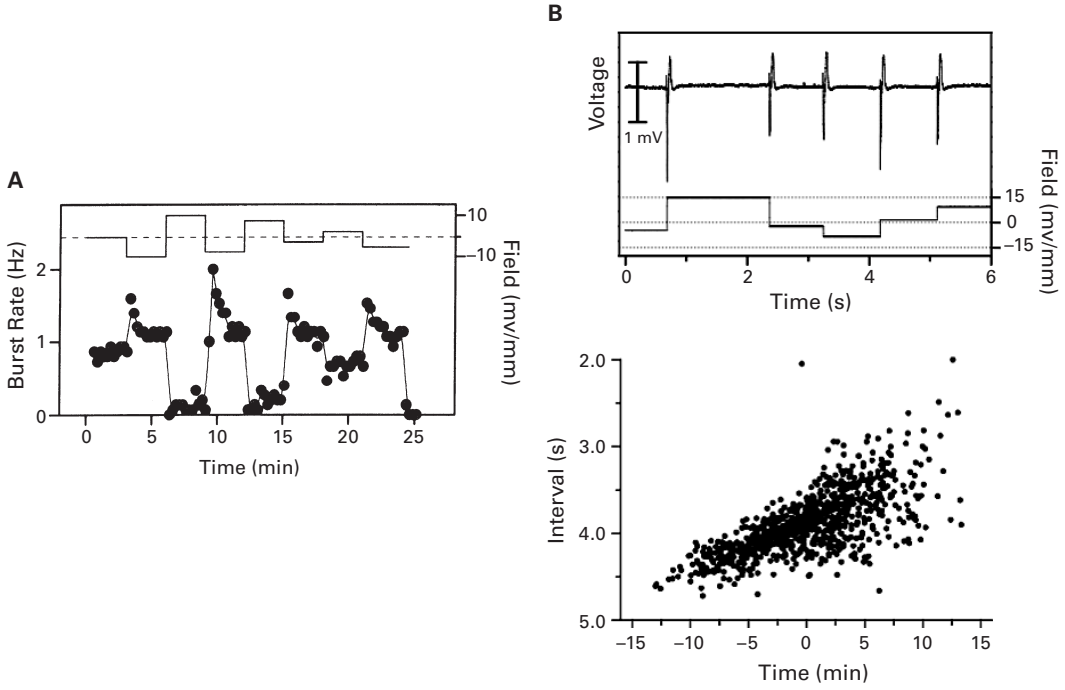


Figure 11.7
A shows how the neuronal ensemble burst firing rate was dependent on the amplitude and polarity of the applied electric field. B shows that even individual burst intervals could be parametrically manipulated using electric field control. A is reproduced from [GNN+96a] with permission. B is unpublished data courtesy of B. J. Gluckman.

the data as multiple near-miss cycles. Look at the time series from chaotic mathematical models—most of them hover near certain values. This is a reflection of such cycles or orbits. Since these orbits have both attracting and repelling regions of phase space where the system will naturally gravitate to or away from such cycles, a simple control scheme can be applied to exploit these inherent dynamics of the system. In figure 11.8, we show examples of both control (C) and anticontrol (D).

We spent years seeking to test the integrity of the orbit structures shown in figure 11.8. We developed a much more rigorous means of extracting such period orbits from experimental data [SOS+96, SOS+97]—only further confirming their presence in neuronal systems. Expanding our search for such orbits in single-cell, small networks, and large-scale networks, we proposed unstable periodic orbits as a novel language for neuronal dynamics [SFN+98].[7]

7. A Chinese translation of this paper appeared in [Qia99].

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

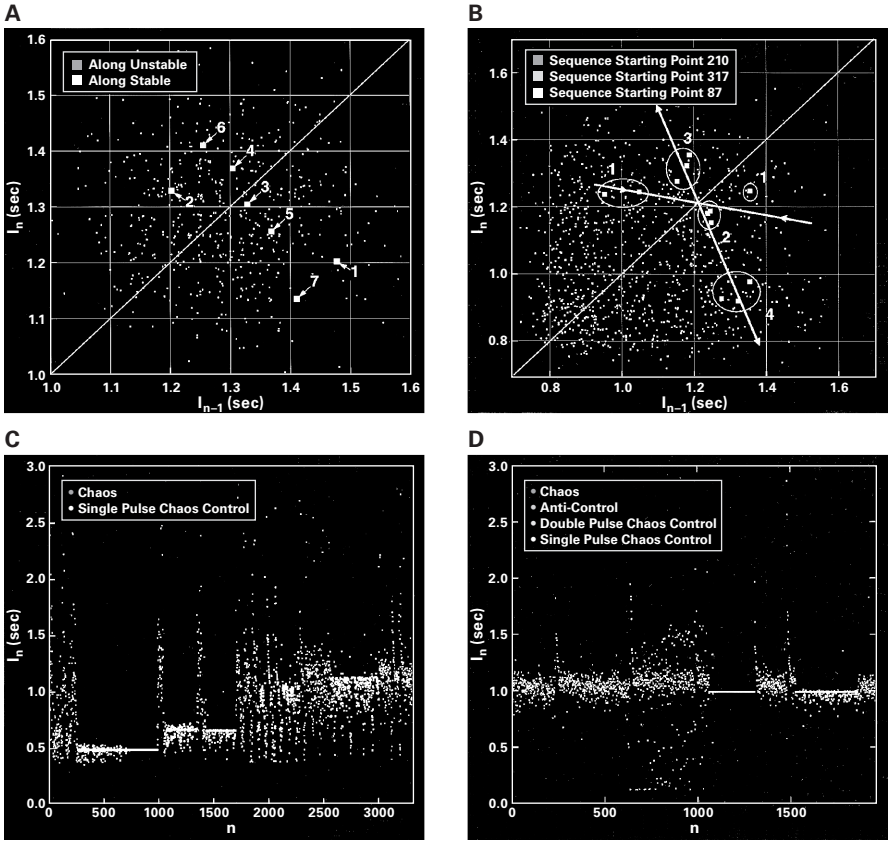


Figure 11.8 (plate 29)
Controlling chaos in the brain. A shows the identification of a motif that shows an approach to the line of identity where sequential intervals, I_n and I_{n+1}, are equal. B shows multiple repetitions of this motif. C demonstrates that by directing the system to fall near the contracting direction (manifold), that the chaotic intervals could be regularized. D shows that by pushing the system away from the contracting direction, that we could prevent regularization. Reproduced from [SJD$^+$94] with permission.

Such chaos control techniques are now known to be a subset [YCX$^+$01] of the more general class of control laws known as variable-state or sliding-mode control [Utk77, DZM88, YUO99]. Armed with electrical field control systems, and with our growing sophistication of sliding mode control strategies, the prospects to fuse these techniques into a much more effective control strategy for neuronal systems is certainly at hand.

A powerful variant of such control ideas for nonlinear systems, where the parameters create a strongly attracting limit cycle, was proposed by Danzl and colleagues [DHM09]. Most neuron models will have such strongly periodic solutions, and they are *not* chaotic. Such solutions often arise concurrently with the creation of an unstable periodic orbit,

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

contained in the phase space *within* the stable limit cycle, the result of a "Hopf" bifurcation as you change a parameter of the neuron model. If you want to desynchronize such a population of neurons, you push it with stimulation to a region where the sets of different phases (termed *Isochrons*) are concentrated, such as an unstable fixed point. In this context, different phases imply different times to the next event or spike. Push a system to a little ball near this unstable point, and noise will randomize the phases of the neurons for the next spike. This control technique can also be used to optimally increase the firing rate by pushing the neuron toward the point on the stable limit cycle corresponding to the spike. The time-optimal nature of the control results in the maximal spike rate increase for a given stimulus magnitude. When applied to a population of neurons, this may result in synchronization with maximal firing rates. My sense is that this type of adaptive control scheme can also be developed from empirical strategies as shown in figure 11.8—an essential step if the rigor and assumptions of [DHM09] cannot be met in practice. Reading this book, you would also wish to consider placing this scheme within a data assimilation framework. The implications for modifying dynamics of conditions such as epilepsy or Parkinson's disease are intriguing.

11.5 And a Fling with Ice Ages

A related experiment possible with such an experimental apparatus was to test the brain for *stochastic resonance*. In stochastic resonance, the response of a nonlinear system to an otherwise subthreshold signal is optimized with the addition of noise [BSV81]. Since its proposal as a mechanism for amplifying the effects of the Earth's small periodic orbital variations by random meteorological fluctuations leading to ice age periodicity [BPSV82], stochastic resonance has been observed in a diverse range of physical systems [WM95]. There was an interesting prediction that in two-dimensional excitable media, spatiotemporal stochastic resonance should be seen in the creation of coherent structures such as spiral waves [JMK95]. Since electrical fields linearly superimpose, we blended sinusoidal signals with random noise, and were able to cleanly demonstrate that stochastic resonance could be observed in neuronal networks [GNN$^+$96b] (figure 11.9). In figure 11.9A, we show the comparison of raw data where random noise is applied through an electrical field to a neuronal network, and below this the combination of random noise and a sinusoidal electrical field. Note that in the presence of the sinusoidal signal, the bursts of the field components tend to occur on the peaks of the periodic signal. In figure 11.9B a family of curves are shown, each at different amplitudes of sinusoid A_{sin}, as a function of noise amplitude A_{noise}. Note that for each amplitude of signal, there is an optimal noise for which the signal is transmitted to the neuronal network. To my knowledge, the two-dimensional spatiotemporal neuronal version of this experiment remains to be done.

Why is stochastic resonance important? Certainly we do not worry about ice age glaciation within the brain. The subtle point is that the *noise* in these examples is a signal that

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

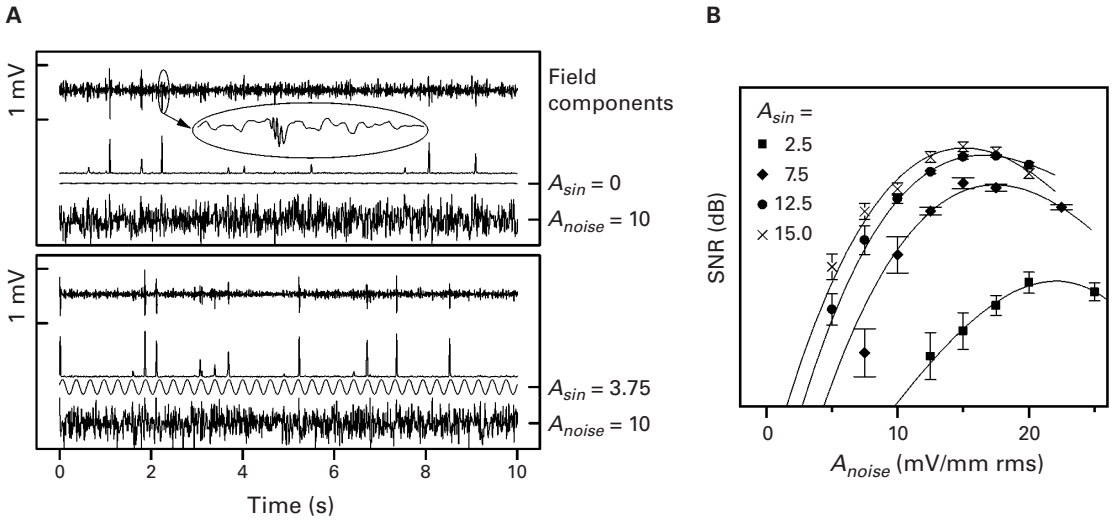


Figure 11.9
A compares between adding noise to a neuronal network through an electrical field (upper) and combining noise with a sinusoidal electrical field signal. Note that, in the combination, the neuronal responses tend to occur at the peaks of the sinusoidal signal. B shows families of curves at a range of sinusoidal signal amplitude A_{sin}. As a function of noise amplitude, A_{noise}, the signal-to-noise ratio (SNR) in decibels (dB) peaks at an intermediate level of applied noise. Reproduced from [GNN$^+$96b] with permission.

is *uncorrelated* with the simple sinusoidal signal at hand. In the brain, the functional result is that *stochastic resonance can occur whenever two uncorrelated signals impinge on neuronal networks*. Now the range of implications explodes: the activity of ongoing inputs to the auditory cortex in the presence of external sounds that are uncorrelated, the multimodal confluence of sensory information processing the lateral interparietal cortex, the list is endless.

Stochastic resonance in a speculative sense is the added energy in ensembles of interacting elements with thresholds that likely fuels neuronal interaction throughout the nervous system. Perhaps the only thing surprising about stochastic resonance would be if it were not found in the nervous system when searched for. One major unsolved question is whether the levels of activity of different parts of the brain are tuned in any way to optimize information flow. Another question that naturally arises is whether the resonance from independent signals promotes enhancement of multiple information streams simultaneously in the same network. Needless to say, the range of implications of this inherent property of nonlinear systems with thresholds is enormous for the nervous system.

11.6 Feedback Control with Electrical Fields

We spent years searching for an effective way to incorporate the electrical field effects from Gluckman and colleagues [GNN$^+$96a, GNN$^+$96b] into a feedback control strategy. There

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

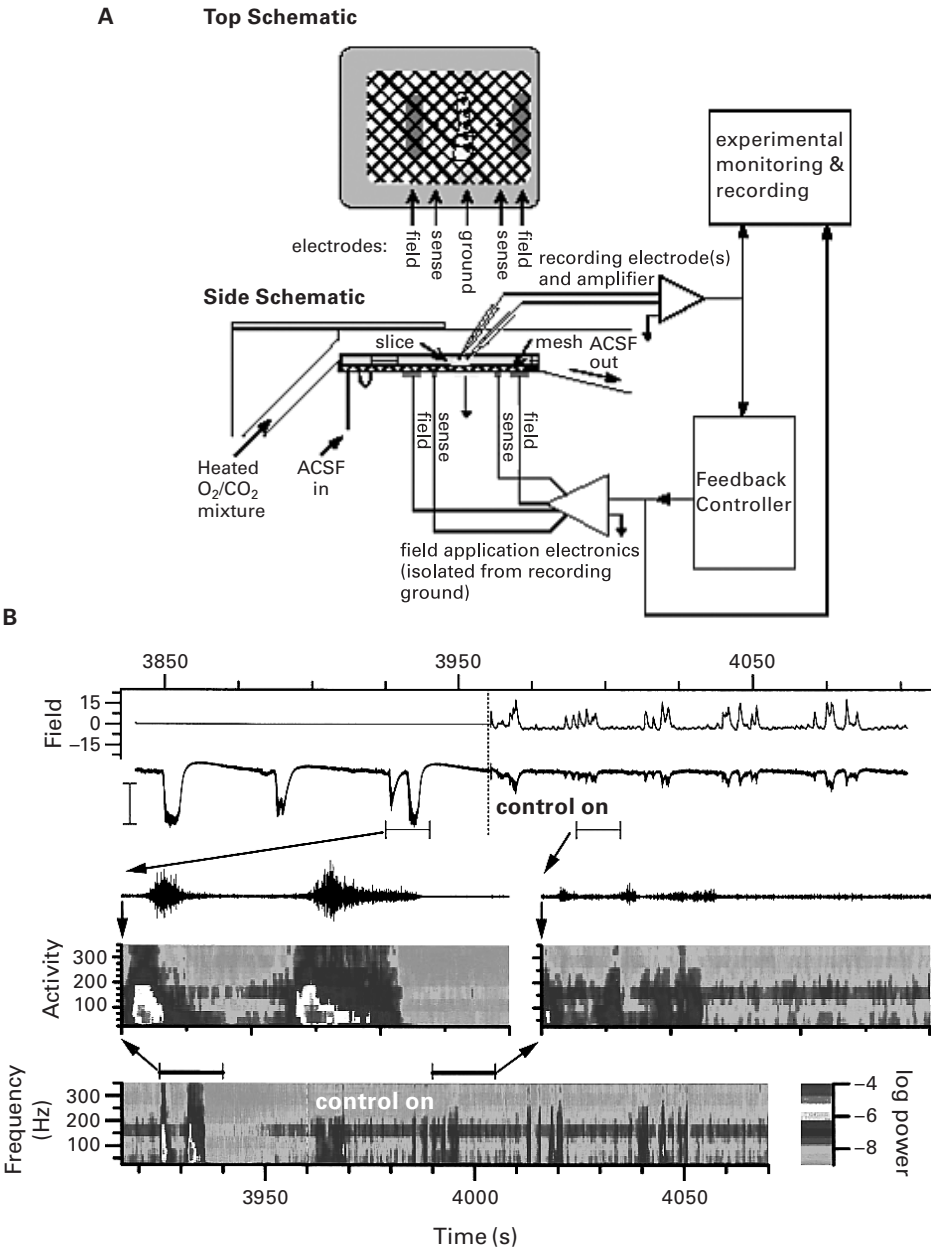


Figure 11.10 (plate 30)
Adaptive feedback control of neuronal activity. Schematic of experimental chamber and electronics shown in A. In B are results demonstrating that a proportional control law could effectively suppress seizure activity in the brain slice. Reproduced from [GNWS01] with permission.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

were some obvious electrical engineering issues that needed to be addressed. We needed to isolate the electronics of the recording system, with an adjustable reference electrode on a field isopotential, from the electronics of the stimulation system, as shown in figure 11.10A (plate 30).

One of our goals was to find a way to suppress epileptic seizures. An advantage of using brain slices is that they preserve the native neuronal architecture from which they are harvested. Interestingly, it was found that exposure of such brain slices to elevated levels of potassium in the perfusate, raising the physiological level of about 3.5 to 8.5 mM [RLJ85], would produce burst firing similar to epileptic spikes.[8] Under certain conditions, such slices exposed to elevated potassium would generate much more prolonged patterns of activity that had a remarkable resemblance to human epileptic seizure patterns [TD88].[9]

It is important to pay attention to the current-generating electrodes that supply the controling electric field. In the experiments in figure 11.10, we used silver-silver chloride electrodes. These electrodes form a very efficient chemical reaction with saline fluids, exchanging chloride ions with solution as charge is passed to the electrodes. But they are toxic if implanted into the brain. For in vitro experiments, where they are used without direct contact with the brain networks, this issue is avoided. But the author must confess that earlier experiments attemped to use stainless steel electrodes, which polarize and do not pass low-frequency currents well. An obvious hallmark of such mistakes is the frequent appearance of gas bubbles around such electrodes—water hydrolyzes at electrode polarizations a bit larger than ± 1 V. Platinum-based electrodes have better electrical characteristics. A very complete review of the often neglected aspects of the materials in such electrodes can be found in Merrill and colleagues [MBJ05].

In figure 11.10B, we show the effect of a simple proportional control law. The control parameter was taken to be the root-mean-square of the high-frequency neuronal activity, from 100 to 500 Hz. The power in such a frequency band was relatively low except during seizures, and was especially prominent at the onset of seizures. With negative gain multiplying this control parameter, and delivering low-frequency (<30 Hz) feedback electrical fields, seizures were reliably suppressed.

Our first efforts employed the rational but unfortunately poor decision to half-wave rectify the control signal. We presumed that we wanted to suppress seizures, and that there was

8. Epileptic spikes are brief intense electrical events seen in the EEG of patients with epilepsy, and only rarely in the EEG of people without a history of epileptic seizures. In EEG terminology, these spikes are less than 70 msec in duration. The burst firing events seen in high potassium are about 100 msec, and have often been presumed (hoped) to be the analogs of epileptic spikes.

9. Producing seizure-like events in elevated potassium is a tricky business. Often Traynelis and Dingledine [TD88] used a period of transient hypoxia to bring out these events. Such posthypoxic hyperexcitability was well described [SS85], but it complicates the relationship of the seizures to the ion manipulations. One can remove the magnesium or calcium from the perfusate, but this seems to the author a drastic physiological approach. A purer "ionic" formulation of reliable seizure generation is to use 4-aminopyridine with a partial reduction in magnesium levels [ZCBS06]. Which is best? No one knows. Such unknowns remind us of the limits to the use of in vitro models for disease analogs.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

only one polarity, with reference to the schematic in figure 11.5, that would accomplish such consistent suppression. This was very successful at suppression so long as the control was maintained. As soon as the controller was turned off, there were intense withdrawal seizures. Our mistake was that this half-wave rectification would have a net polarization effect on the brain region in bulk. We suspect that building up potassium asymmetrically within the slice or layers was likely a culprit [GMN83].

By employing a balanced charge (nonrectified) proportional-integral control law, these withdrawal seizures were eliminated. We calculated the net current output over the most recent 0.5 to 1.5 sec of controller operation, and then compensated for this recursively in our controller output. The full analysis of proportional versus proportional-integral control can be found in Gluckman and colleagues [GNWS01].

Such difficulties in applying control engineering principles to control of neuronal circuits emphasizes the need for animal experiments to test even the most logical of ideas. In chapter 12, we will discuss ways of fusing models of the dynamics and diffusion of potassium ions in brain tissue with compartmental neuronal models expressing Hodgkin-Huxley dynamics. The more refined our models become, the more we hope to minimize the need for animal experiments in what should be the rational design of neuronal controllers. But in the end, testing such controllers on something as complex as the actual brain is, for the foreseeable future, a necessity. Unlike Boeing and their 777 airframe, we still need to test fly the designs of even the simplest of brain control systems to see if they crash.

11.7 Controlling Propagation—Speed Bumps for the Brain

If one pharmacologically blocks the fast (GABAa) inhibition in brain, then small stimuli can produce traveling waves across the cortex. They tend to travel (or perhaps are led by) the middle cortical layers such as layer 5 [TC98].

As mentioned in chapter 6, the Wilson-Cowan equations can produce traveling waves. Such nonlinear wave phenomena are common in both physics and biology, and it was noted some time ago that inhomogenous "bump" type solutions of even stationary waves would characteristically have a spatially larger stable and a spatially smaller unstable solution [AMa77]. Intuitively, one needs, as with nonlinear waves in general [Sco03], enough regenerative energy supplied to offset the tendency of such waves to dissipate. In Wilson-Cowan equations, this requires a sufficient amount of "input" from other regions of the network to keep the core spatial region sufficiently activated. In figure 11.11A, we show a traveling wave solution $u(x, t)$ for Wilson-Cowan equations, with threshold θ and propagation speed C. Figure 11.11B shows the characteristic plots of threshold θ versus propagation speed C. There are two solutions along each branch—the upper one is stable, and the lower one (within the gray region) is unstable.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

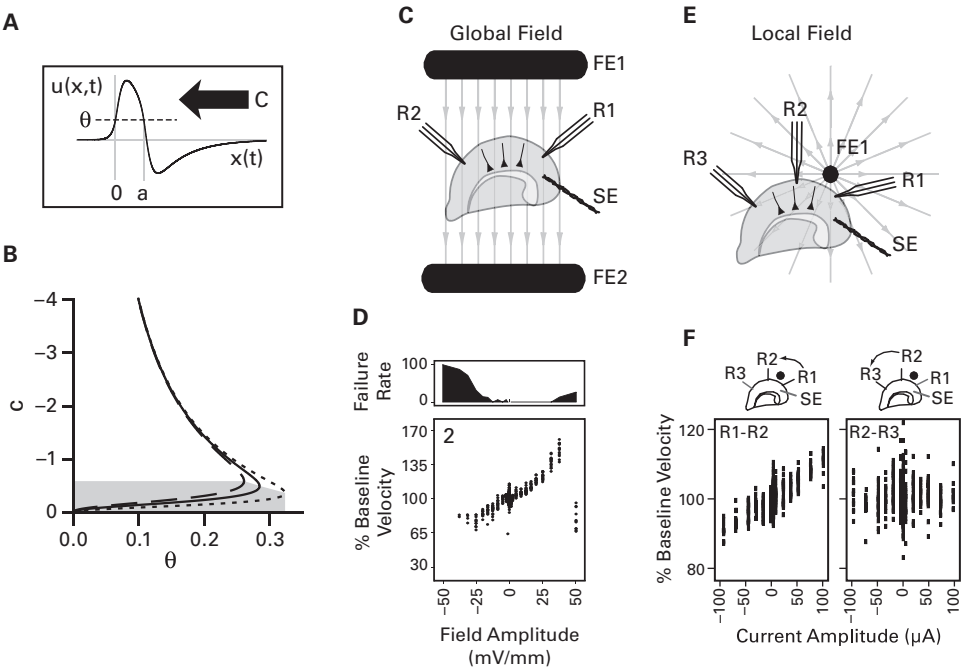


Figure 11.11
Traveling waves in cortex, theoretically and experimentally. A shows a schematic of a traveling wave solution, where we plot a traveling wave solution $u(x, t)$ for Wilson-Cowan equations, with threshold θ and propagation speed C over distance x. B shows the characteristic relationships for velocity C versus threshold θ. C shows a schematic of an experiment where global electrical fields are applied, between field electrodes FE1 and FE2, to waves initiated by stimulating electrode SE, and recorded with recording electrodes R1 and R2. D shows the modulation of velocity with applied field amplitude, and the failure rates of initiated waves. E shows the application of radial electrical fields, and F shows the modulation of velocity with only close proximity to the punctate field source. Reproduced from [RSG05] with permission.

Figure 11.11C shows the experimental setup whereby a global electrical field is imposed on a cortical brain slice. A traveling wave is initiated by the stimulating electrode (SE) and travels first by recording electrode 1 (R1) and then reaches recording electrode 2 (R2). The wave velocity, C, can be increased or decreased by alternating the field polarity about zero, as shown in figure 11.11D. As theory predicts in figure 11.11B, when the waves are slowed to a sufficiently small value of C, they become unstable, which is shown as an increasing failure rate in figure 11.11D.[10]

10. Note that at high field strengths where velocity is peaked, there are also some increasing propagation failures. Modeling suggests (unpublished data) that such failures may be related to the increasing hyperpolarizing blockade that will develop in the dendrite fields of asymmetric neurons when the somata are depolarized. Pursuit of this conjecture requires further study.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

Now note the effect of a radial electrical field produced by a punctate stimulating electrode made from a pellet of silver-silver chloride, shown in figure 11.11E. When the waves pass by close to the source of these punctate field sources, the propagation velocity can be increased or decreased by alternating the field polarity about zero, as shown in figure 11.11F. When the wave passes between electrodes a short distance away, its velocity is unaffected by the field. Blockade of these waves is difficult to produce with such punctate field sources, perhaps reflecting inhomogeneity in the field, or transmitting fibers coursing *en passage* through the region of the most intense field polarization. For traveling waves, such punctate fields act less as brick walls than as speed bumps.

In terms of controller design for seizure activity, one would ideally predict and prevent a seizure from initiating. Seizure prediction at the time of this writing remains more a goal than reality [LL05, MKR$^+$05]. The next goal would be to rapidly interact with a seizure to stop it, as shown in figure 11.10. If a seizure escaped being crushed early on, one could implement a third strategy—blocking propagation, as shown in figure 11.11. One can, of course, implement all three goals in a tiered device design. We envision that placing both a set of electrodes at a seizure focus, in an effort to predict and disrupt, and then a more distant set of electrodes designed to cut off propagation if the first strategies fail would be a sensible design for such a controller and electrode deployment. For seizure-prone regions of the brain, such as the hippocampus, this strategy can be employed with a single electrode shaft, with contacts at the tip of the shaft targeting the epileptic focus, and electrodes more distal from the tip available for propagation blockade. Such axial electrodes are in common use in clinical practice and can be targeted precisely from an entrance site at the occiput using image-guided stereotactic surgical placement. Such minimally invasive approaches to control electrode deployment drive our configuration for the translational in vivo experiments discussed in more detail later.

11.8 Neurons in the Resistive Brain

Computational neuroscience has, not surprisingly, tended to focus on neurons [KS98, Dal01]. But as the previous material has been leading us to conclude, one needs to consider embedding our model neurons within a resistive lattice that represents the electrical characteristics of the brain. As should also be obvious, we also need to take into account the physics of lower-level ionic dynamics representing the transmembrane pumping, cellular buffering (especially by glial cells), and diffusion of ions that are intimately linked to determining excitability. We will address metabolic ion dynamics in considerably more detail in chapter 12, but first let's examine the resistive brain.

The pioneering computational work on the electrical field interactions between neurons through their endogenous fields was performed by Roger Traub and colleagues [TDTK85, TDSK85]. It is essential to use multicompartmental neuronal models to represent

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

polarization effects. Whereas Traub and colleagues focused on multicompartmental representations of dendrites, soma, and axons [TM91], we have already seen how Tranchina and colleagues [TN86] reduced such models to equivalent cylinders because of the symmetries inherent in polarizing symmetrical dendrite fields. Pinsky and Rinzel [PR94] demonstrated how much of the qualitative properties of multicompartment pyramidal cell firing could be represented in a simpler two-compartment model where the properties of dendrites and soma could be lumped into single composite compartments. The Pinsky-Rinzel model is the minimally reduced model that can represent polarization effects from an applied electrical field.

In Park and colleagues [PBG$^+$05], we embedded Pinsky-Rinzel two-compartment neurons (figure 11.12A) within an electrically resistive matrix (figure 11.12B). We created equivalent terminal resistances (z) to represent the equivalent networks of a resistive matrix of (for all the neuron cares) infinite extent. We coupled these neurons, only excitatory here, with two types of excitatory synapses.

We were keen to study the effect of applied electrical fields on neuronal syncyhrony. By varying the nature of the cells (the internal conductances), and the natural frequency of their spiking, we were able to study the degree of synchronization as a function of applied electrical field and neuronal disparity. The results demonstrated a complex landscape between synchrony and asynchrony, as shown in figure 11.12C. Especially interesting were the phase relationships that emerged from a study of the influence of the relative phase difference between cellular firing and the response of the neighboring cells. In contrast with the phase response curves dependent on synaptic connectivity (figure 11.12Da, Dd), the phase response curves as a function of the electrical ephaptic effects alone were of considerably smaller magnitude (figure 11.12Db, Dc). Nevertheless, at particularly sensitive phases, the ephaptic effects were important. In this modeling, we incorporated a simple model relating potassium extrusion from activity in cells to a change in the proportionality of intracellular to extracellular electrical resistance.

11.9 How Small an Electrical Field Will Modulate Neuronal Activity?

As discussed previously [TB56], the electrical field strength required to modulate neurons that are actively firing is much lower than that required to fire neurons from rest. How small an electrical field will do this?

Theoretical calculations for elongated cells exposed to an electrical field predicted that fields as small as 100 μV/mm would be a lower bound on such effects [WVAA98]. The key here was that an external electrical field needed to overcome the *molecular shot noise* produced by biochemical processes within cells [WVAA98]. It was clear from experiments that the limit of interaction was lower than 1 mV/mm [GBD00], but the discrepancy between theory and experiment remained.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

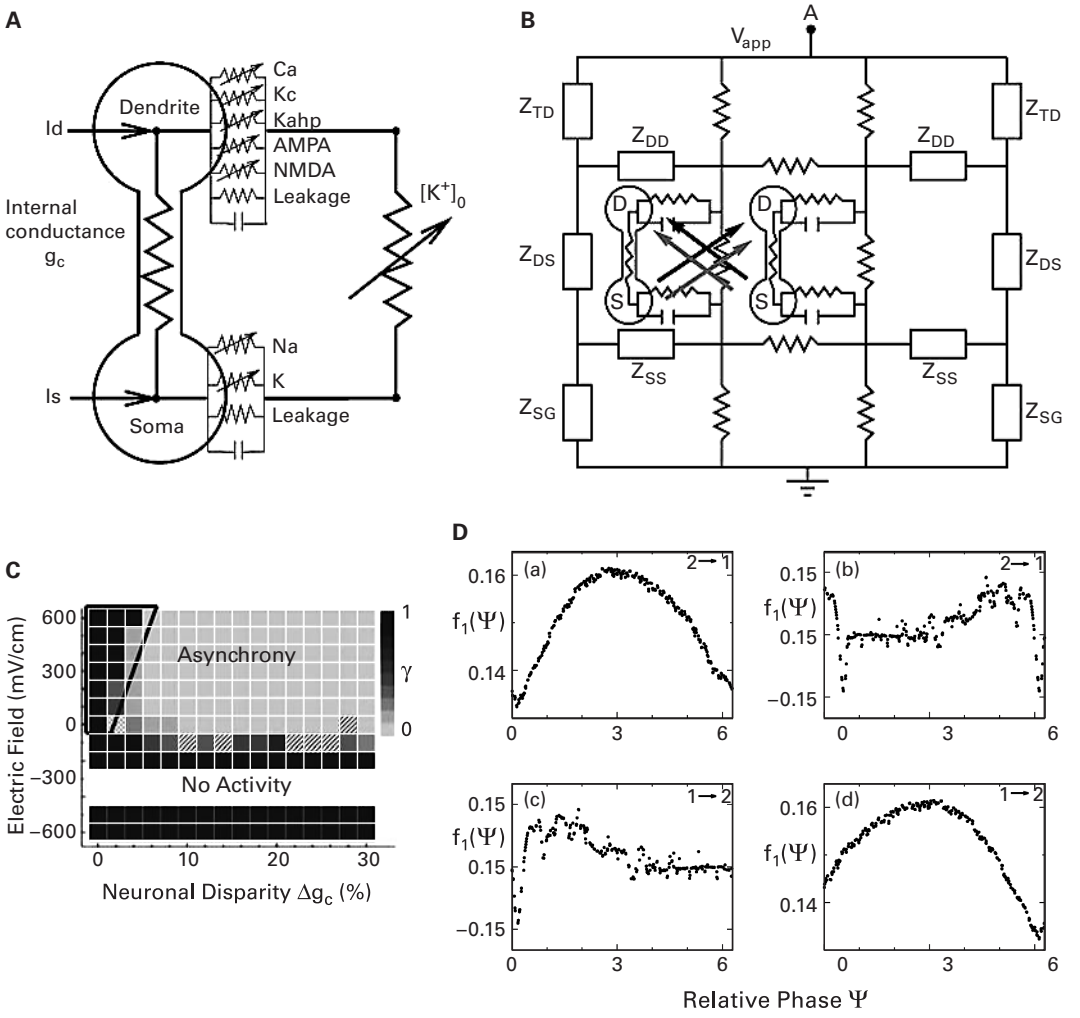


Figure 11.12
A shows the Pinsky-Rinzel lumped two-compartment model of a pyramidal cell, and the relevant conductances and membrane currents. In B, these neurons were embedded within an electrically resistive array with terminal resistances to represent an array of infinite extent in comparison with the dimensions of the neurons. C shows the complex interplay between applied electrical field, neuronal disparity, and 1:1 synchrony (solid black). Gray scale γ reflects intensity of synchronization. The hatched boxes represent higher-order phase locking beyond 1:1. D shows numerical simulations of the phase response curves of each neuron influencing the neighboring neuron from synaptic (a,d) and ephaptic electrical field (b,c) effects. The analytical derivation of these curves is outlined in detail in [PBG$^+$05].

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

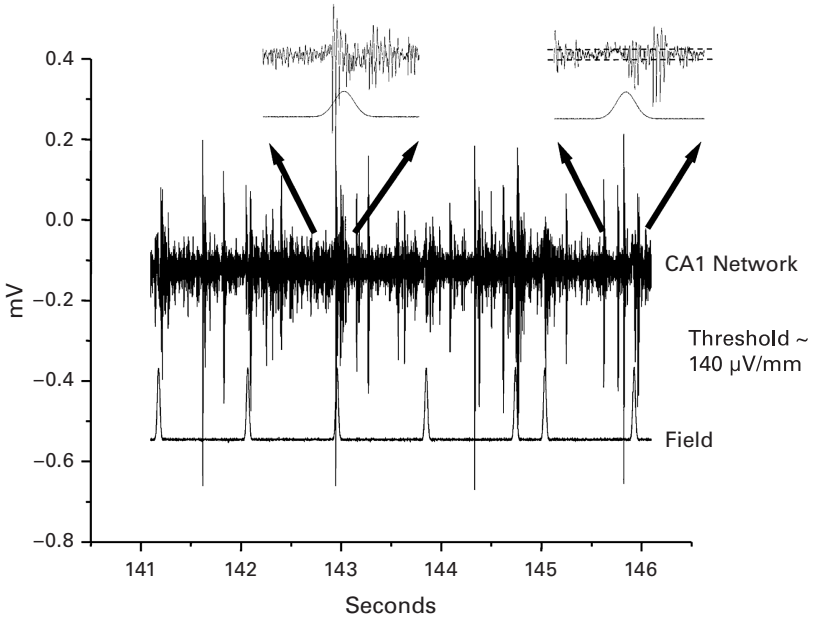


Figure 11.13
Experimental search for the electrical field threshold to modulate neuronal firing. We employed small field pulses with Gaussian waveform, to emulate naturally occurring burst firing activities. On the CA1 network, the insets show the raw data illustrating modulation of ensemble firing in response to Gaussians with a 140 μV/mm root-mean-square amplitude shown above. Careful statistics, including perturbations in the timing of the Gaussian pulses, demonstrate that modulation at this amplitude is statistically significant. Reproduced from [FGS03] with permission.

We sought to explore this lower limit [FGS03]. In figure 11.13, we used an electrical field similar to that of burst firing within a nervous system. Such pulses of intense neuronal activity can be seen within neuronal networks during active exploration, for instance [BUZ86], where local field intensities of over 10 mV/mm can be measured within the brain. We demonstrated that the smallest such pulsed field that can modulate neuronal firing was 140 μV/mm root-mean-square, 295 μV/mm peak amplitude; this validated the theoretical predictions. We further showed that these networks are more sensitive than the average single neuron threshold (185 μV/mm root-mean-square, 394 μV/mm peak). The origin of the enhanced network sensitivity might well reflect array enhancement, perhaps through a stochastic resonance effect [LMD$^+$95, Gai00, KSK00].

These findings have several implications of importance. First, the strength of the electrical field required to modulate the firing frequencies of elongated and asymmetrical neurons is very small. It is about 2 orders of magnitude less than the strength of the electrical fields generated in the course of normal functioning of the brain [BUZ86]. So there is little question that the brain synchronizes its neurons in part through electrical field effects of its own

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

neurons [AMB+10]. The functional implications of this remain unclear. Certainly during population events, such as navigational theta (4 to 12 Hz in rodents; see chapter 7) rhythms, the phase relations of individual neuronal firing to the population may well be modulated at least in part through endogenous electrical field interactions. But these findings also raise the spectre of the possible public health effects of exogenous environmental electrical fields, which has been a subject of intense controversy [JDBF03, SJ07].

Environmental electrical fields vary widely in frequencies. Our cell phones produce megahertz-gigahertz frequencies—far faster than the polarization time constants of neurons. Such frequencies may affect macromolecular bonds and nucleic acids, and the scientific understanding for such radiofrequency radiation remains a subject of controversy [FAK05, Woo06]. Cell phone frequencies may cook molecules within your cells, but such interactions are useless for our focus here on neural system control design. One of the problems in the environmental assessment of electrical fields is that the actual measurements of field attenuation by scalp, skull, and the coverings of the brain has often been lacking in our assessments [Ada91]. More exacting science, whether at radiofrequencies, or at the extremely low frequencies in the ranges where modulation of neuronal firing is expected, seems warranted.

11.10 Transcranial Low-Frequency Fields

The thresholds for electrical field interaction with brain seem so low [FGS03] that the bold might attempt some truly strange experiments. The application of slow transcranial potentials (0.75 Hz) across the scalp at small current densities during sleep as performed by Marshall and colleagues, [MHMB06] could enhance the retention of declaritive memories in human volunteers. In a comparable experiment during wakefullness [KWS+09], the encoding of memories was enhanced during learning. Interpretions of such experiments leaves many unanswered questions at present. Yet recent experiments suggest that both slow oscillatory and constant transcranial DC stimulation can alter excitability of motor cortex [BGS+09]. Recent experiments have demonstrated that similar transcranial electrical fields in animals can indeed alter the firing frequency of neurons [OSB+10]. Whether such slow or steady fields work through polarization, as discussed in this chapter, or also involve ionic shifts, as we will discuss in more detail in the next chapter, remains an exciting area for future research.

11.11 Electrical Fields for Control Within the Intact Brain

The goal of this chapter was to get to this section. Can we apply the principles of closed loop control to directly interact with the dynamics of neurons within the brain? And can we attempt to modulate such neurons to specifically target pathological dynamics such as epileptic seizures?

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

From the preceding discussion, we see several parts of the mammalian brain where the neuronal architecture is such that the large outflow excitatory neurons, often termed *pyramidal cells* for their shape and more generally labeled as *principal cells*, are aligned in parallel formation. The hippocampus has the principal cells within the *cornu ammonis* aligned as such, although the curvature of this structure presents some engineering issues in terms of optimal electrode configuration, as shown in figure 11.14.

We spent considerable time, following Gluckman and colleagues [GNWS01], attempting to implant small platelike electrodes to replicate the arrangement of polarization electrodes, as shown in in vitro experiments illustrated in figure 11.6. There was sufficient damage to the structures connecting to the hippocampus to render such implants useless, regardless of perceived surgical skill. We then explored a series of experiments using an axially placed wire electrode that coursed through the long central axis of the hippocampus. Such experiments were motivated by our clinical experience with such axial electrodes, where their use in the localization of epileptic seizure foci is a common approach [HSW+91]. An example of such a posterior placement of an axial depth electrode is illustrated in figure 11.15A (plate 31). In our experiments, we prepared steel electrodes with surfaces of electrodeposited thin

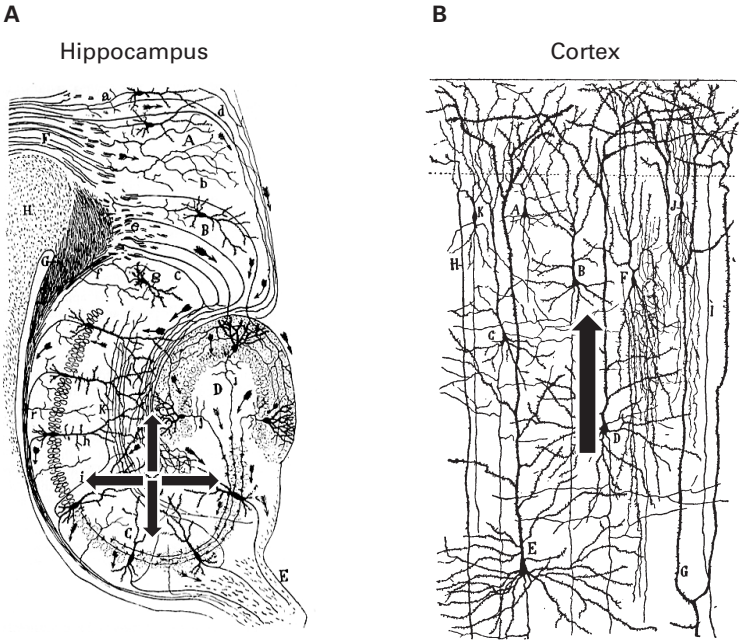


Figure 11.14
A shows image of the hippocampus. An added radial electrical field source is illustrated as arrows within the curve of the *cornu ammonis*. Reproduced from [Caj09]. B illustrates cortex, again demonstrating field alignment as an arrow. Reproduced from [Caj09].

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

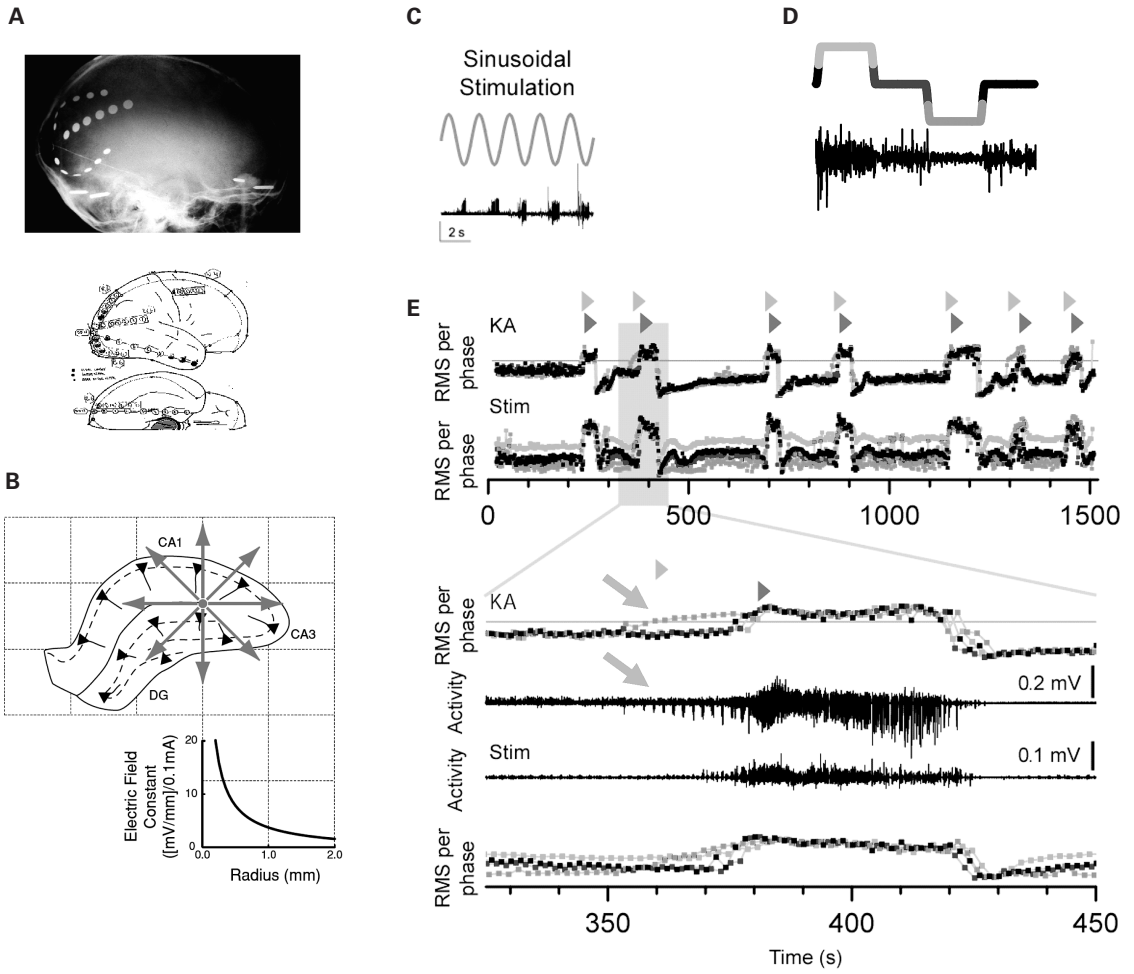


Figure 11.15 (plate 31)
A shows an example of occipitally placed recording depth electrode in human epilepsy patient. Reproduced from [SSKW05] with permission. B is schematic of an axially placed stimulating electrode for epilepsy control in experimental hippocampal seizures. C shows experimental verification in animal models of sinusoidal entrainment of hippocampal neurons with low-frequency sinusoidal electrical fields. Using DC electrical fields of alternating polarity, in D, we demonstrated in E that by stimulating the contralateral side to the kainic acid (KA)–injected side with subthreshold alternating pulses, that the network demonstrated progressively increasing responses to these small stimuli in the seconds leading up to the start of the seizure (red arrows). B through E are reproduced from [RGW$^+$03] with permission.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

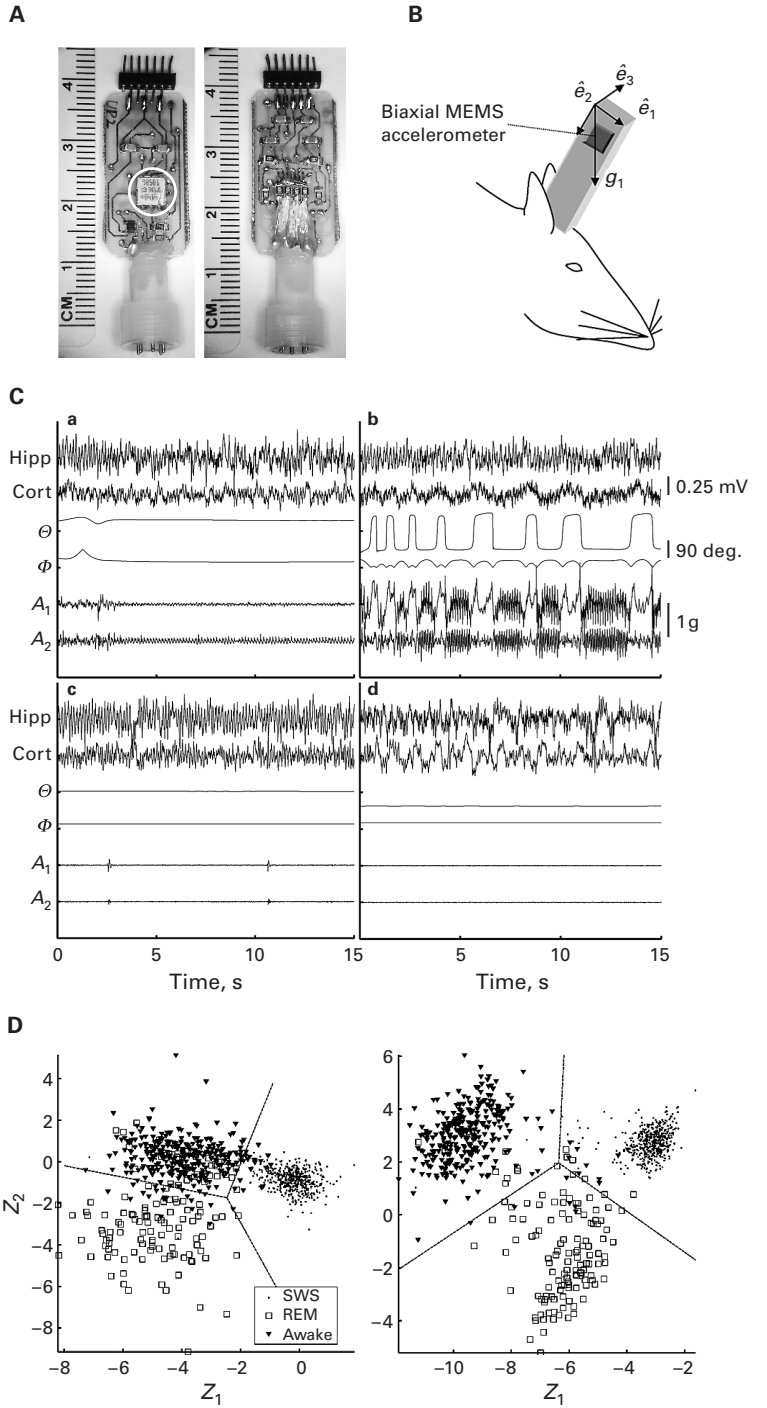


Figure 11.16
(Caption on facing page)

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

films of iridium oxide [MCNR01]. These surfaces were designed to minimize the polarization and maximize the charge-carrying capacity of such stimulation electrodes. This is critical to prevent the polarization of the electrode surfaces to the extent that irreversible chemical reactions take place at the electrode surface [MBJ05], and to maximize the amount of charge that can be safely passed while constraining the potential developed to within strict safely limits [MBJ05].

We could thus generate a radial electrical field, as illustrated in figure 11.15B [RGW+03]. It was apparent, in these experiments, that we could sinusoidally entrain hippocampal neurons as shown in figure 11.15C. We then performed a series of alternating DC pulse stimuli (figure 11.15D) opposite to the side of the kainic acid–induced acute epileptic focus, and recorded from the side ipsilateral to the focus. We used pulses that were subthreshold to stimulate the neuronal circuitry in the interictal (between seizure) state. One of our findings was that, in the tens of seconds prior to a seizure, we could detect an increasing amplitude of fluctuations from the neuronal network, as shown in figure 11.15E. Whether such weak dynamical *probes* of epileptic networks could be employed to increase the accuracy of seizure prediction is a question that warrants further exploration.[11]

11.12 To Sleep Perchance to Dream

Absent from this discussion, and from most of the literature on EEG dynamics and seizure prediction, is the issue of sleep. Nevertheless, the *state of vigilance* of an animal or human has a dramatic effect on both behavior and EEG dynamics. Our present quantification of state of vigilance tends to be an amalgam of EEG, electromyography, and video recording of a person's behavior. Electromyographic electrodes are uncomfortable—they impale your muscles. It is infeasible for an automated control device to rely on video interpretation of wakefulness.

We have accordingly explored the use of microelectromechanical (MEMS) accelerometers as an adjunct to EEG recordings [SCP+07]. An example of the integration of MEMS accelerometers into an electronic head stage is shown in figure 11.16. Panel B is a schematic of

11. One would certainly like to speculate that seizures might be analogous to phase transitions in physical systems [SRSRSL99]. In such systems, near the critical points of phase transitions, the fluctuations within ensemble systems can increase [Sta87]. Attractive as this notion might be, it is too speculative at present to warrant more than a footnote in this text.

Figure 11.16
Incorporation of MEMS accelerometer into head stage EEG apparatus in A, and schematic of use in B. C shows examples of various behaviors, reflected in hippocampal electrode (Hipp), cortical electrode (Cort), steady (DC) head tilt angles θ and ϕ, and dynamic (>1 Hz) biaxial acceleration plotted in A_1 and A_2. D shows linear discriminant analysis without (a) and with (b) acceleration data included in the discriminator. Z_1 and Z_2 represent discrimination variable (optimal linear combinations of original variables). Reproduced from [SCP+07] with permission.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

the head acceleration data measurements. The types of signature of EEG and accelerometer data in drinking, grooming, rapid eye movement sleep, and slow wave sleep, are shown in panel Ca–d, respectively. By creating a linear discrimination classifier for these combined accelerometer and EEG data [Flu97, SSKW05], we were able to show that the addition of acceleration data improves the state of vigilance classification of many of the presently used sleep discrimination algorithms [SCP+07].

No epilepsy control device, and perhaps no sophisticated Parkinson's disease controller, can be designed without taking state of vigilance discrimination into account in the algorithms.

11.13 Toward an Implantable Field Controller

The next stage in the development of an electrical field controller is to design it to function in awake behaving animals. Can we interact with seizures in open loop in this setting?

In [SCP+09] we used an axial hippocampal trajectory, which in people would be placed from an occipital approach, but in rodents a vertical approach is easier, as shown in figure 11.17. Making certain that the response of the neuronal responses in such settings is not caused by stimulation artifact is a complex task. We calculated a transfer function, α, relating the applied current to the electrical potential as

$$V(t) = \alpha \otimes *I(t)$$

where $\otimes$ denotes convolution. Ideally, one compensates for frequency-dependent amplitude and phase of a transfer function such as this [CSG08]. For now, we will assume that with periodic stimuli, α is constant with zero phase lag. We assume that the recorded signal, $X(t)$, is a summation of the neuronal signal, $N(t)$, with the voltage created by the applied current as

$$X(t) = N(t) + \alpha I(t)$$

If $N(t)$ and $I(t)$ are uncorrelated, then

$$\alpha = \langle X(t)I(t)\rangle \Big/ \left\langle I^2(t)\right\rangle$$

where $\langle\cdot\rangle$ indicates time average. Artifact is then removed from the signal, leaving an estimated neuronal signal $\hat{N}(t)$ as

$$\hat{N}(t) = X(t) - \alpha I(t)$$

Using such an artifact-reduced signal, we show spectrograms using multitaper spectral estimates [Tho82, PW93] in figure 11.17Ba and Be for an unstimulated and stimulated seizure, respectively. In this particular type of tetanus toxin seizure [FJ00], the waveforms

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

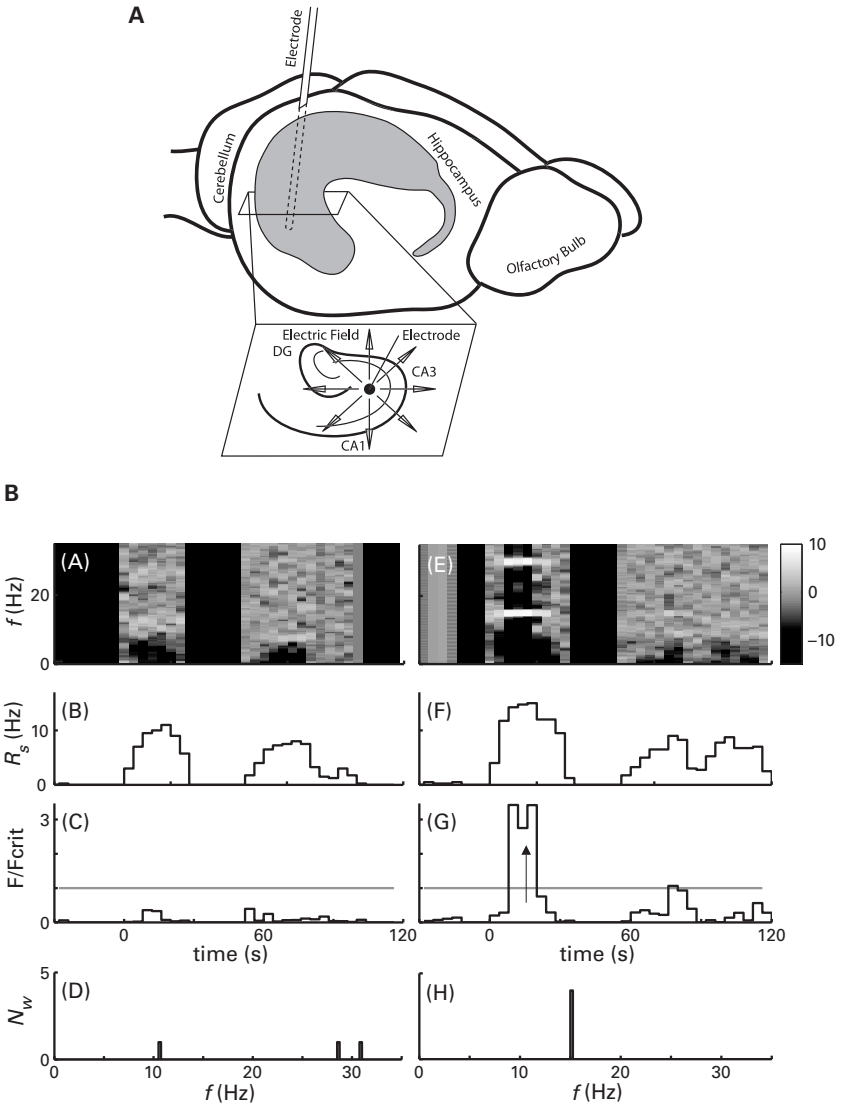


Figure 11.17
A shows a schematic of axial placement of hippocampal stimulating electrode within the ventral axial hippocampus. B shows examples of seizures with and without low-frequency electric field stimulation. Both seizures were taken from the same animal, recorded from a depth electrode in the same hippocampus as the stimulation. Analysis was done for 150 sec starting 30 sec prior to seizure onset. The seizure on the left (Ba–c) was unstimulated, and the one on the right (Be–g) was stimulated at 15 Hz. Analysis was done in 4-sec nonoverlapping windows. Shown are (Ba, Be) spectrograms computed from the spike-time series, (Bb, Bf) average spike rate R_s per 4-sec window, (Bc, Bg) Thomsons F-statistic normalized to a critical value (0.1%) at 15 Hz. No obvious persistent spectral lines are observed from the unstimulated seizure, and the observed values are consistent with chance. The stimulated seizure has a significant line at 15 Hz that persists for four windows, which is highly unlikely ($p \sim 10^{-7}$) from chance. Reproduced from [SCP+09] with permission.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026

can be decomposed into spikes that are readily assigned times as point processes. These spike rates are plotted in figure 11.17Bb and Bf. Because we are stimulating here periodically, the spectral analysis would show a single spectral line resulting from the stimulation, and we can employ Thomson's F statistic [Tho82] to test whether the amount of power observed at that line of interest is more than we expect by chance after the artifact has been removed. The confidence line for this F statistic is shown in figure 11.17Bc and Bg, demonstrating that there is a very significant entrainment of this seizure at the stimulation frequency.

So we can entrain seizures. The next step of creating feedback control laws to intelligently interact with such seizures is presently underway.

The use of low-frequency electrical fields to modulate neuronal firing is not well suited to different parts of the brain. What we have described here is best suited to regions of the brain where neurons—perhaps only the large outflow neurons such as pyramidal cells—are all oriented in a fashion that allows us to devise electrical fields to align with the long axis of the neurons. Such neurons need to be asymmetrical, so that more radially configured neurons, like many of the inhibitory neurons (or at least their dendrite fields), are less affected. The hippocampus is a region of the brain where the pyramidal cells are generally aligned in a curved sheet so that a radial electric field as we have presented here will effectively modulate the neurons. In cortex, neurons are again oriented vertically so that their somatodendritic axes point normal to the cortical surface. In primates, the curvature of the gyri and sulci of the folded cortex complicate this alignment, and cortical stimulation in the manner of Marshall and colleagues [MHMB06] will tend to preferentially polarize only a subset of the surface area (experiments such as [OSB$^+$10] were performed in lissencephalic rodents, where there are no sulci to interfere with field effects). At present, it is not at all clear (although in fairness, it has not been explored either) whether stimulation of deep nuclei in Parkinson's disease can take advantage of such polarizing fields. Although chapter 10, on Parkinson's disease, focused on pulse stimulation, a separate line of research explores the electrical field interaction from such deep brain stimulation pulses [MGST04].

Further work on the interaction of electrical fields, whether low-frequency continuous or pulsed, is very much warranted in order to understand how to improve controller design.

Downloaded from direct.mit.edu/books/book/chapter-pdf/224785/9780262312080_cas.pdf by Eindhoven University of Technology user on 06 November 2026